
CHAPTER 3

Two ways statistics

In a lot of statistical research, we are not interested in just one character but several at the same time. When we study two characters X and Y on a given population, it is generally because we want to know if there is a link between them and what is the intensity of the link.

Example of possible relationships between the following variables: height and age; diabetes and weight, cholesterol level and diet, ecological niche and population, sunshine and plant growth, toxin and metabolic reaction, survival and pollution, effects and doses...etc. The characters studied can be both qualitative and quantitative.

Correlation is a statistical technique to ascertain the association or relationship between two or more variables. Correlation analysis is a statistical technique to study the degree and direction of relationship between two or more variables. A correlation coefficient is a statistical measure of the degree to which changes to the value of one variable predict change to the value of another. When the fluctuation of one variable reliably predicts

a similar fluctuation in another variable, there's often a tendency to think that means that the change in one causes the change in the other.

Uses of correlations:

1. Correlation analysis helps in deriving precisely the degree and the direction of such relationship.
2. The effect of correlation is to reduce the range of uncertainty of our prediction. The prediction based on correlation analysis will be more reliable and near to reality.
3. Correlation analysis contributes to the understanding of economic behavior, aids in locating the critically important variables on which others depend, may reveal to the economist the connections by which disturbances spread and suggest to him the paths through which stabilizing forces may become effective
4. Economic theory and business studies show relationships between variables like price and quantity demanded, advertising expenditure and sales promotion measures etc.
5. The measure of coefficient of correlation is a relative measure of change.

Types of Correlation:

Correlation is described or classified in several different ways. Three of the most important are:

- I. Positive and Negative
- II. Simple, Partial and Multiple
- III. Linear and non-linear

I. Positive, Negative and Zero Correlation.

Whether correlation is positive (direct) or negative (in-versa) would depend upon the direction of change of the variable.

Positive Correlation: If both the variables vary in the same direction, correlation is said to be positive. It means if one variable is increasing, the other on an average is also increasing or if one variable is decreasing, the other on an average is also decreasing, then the correlation is said to be positive correlation. For example, the correlation between heights and weights of a group of persons is a positive correlation.

Negative Correlation: If both the variables vary in opposite direction, the correlation is said to be negative. It means if one variable increases, but the other variable decreases or if one variable decreases, but the other variable increases, then the correlation is said to be negative correlation. For example, the correlation between the price of a product and its demand is a negative correlation.

Zero Correlation: Actually it is not a type of correlation but still it is called as zero or no correlation. When we don't find any relationship between the variables then, it is said to be zero correlation. It means a change in value of one variable doesn't influence or change the value of other variable. For example, the correlation between weight of person and intelligence is a zero or no correlation

II. Simple, Partial and Multiple Correlation:

The distinction between simple, partial and multiple correlation is based upon the number of variables studied.

Simple Correlation: When only two variables are studied, it is a case of simple correlation. For example, when one studies relationship between the marks scored by student and the attendance of student in class, it is a problem of simple correlation.

Partial Correlation: In case of partial correlation one studies three or more variables but considers only two variables to be influencing each other and the effect of other influencing variables being held constant. For example, in above example of relationship between student marks and attendance, the other variable influencing such as effective

teaching of teacher, use of teaching aid like computer, smart board etc are assumed to be constant.

Multiple Correlation: When three or more variables are studied, it is a case of multiple correlation. For example, in above example if study covers the relationship between student marks, attendance of students, effectiveness of teacher, use of teaching aids etc, it is a case of multiple correlation.

III. Linear and Non-linear Correlation:

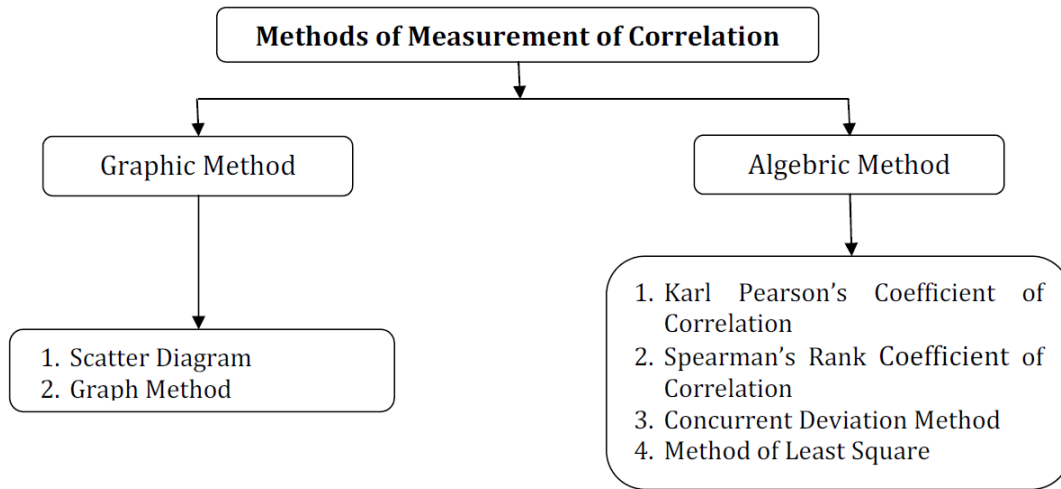
Depending upon the constancy of the ratio of change between the variables, the correlation may be Linear or Non-linear Correlation.

Linear Correlation: If the amount of change in one variable bears a constant ratio to the amount of change in the other variable, then correlation is said to be linear. If such variables are plotted on a graph paper all the plotted points would fall on a straight line. For example: If it is assumed that, to produce one unit of finished product we need 10 units of raw materials, then subsequently to produce 2 units of finished product we need double of the one unit.

Non-linear Correlation: If the amount of change in one variable does not bear a constant ratio to the amount of change to the other variable, then correlation is said to be non-linear. If such variables are plotted on a graph, the points would fall on a curve and not on a straight line. For example, if we double the amount of advertisement expenditure, then sales volume would not necessarily be doubled.

Methods of measurement of correlation:

Quantification of the relationship between variables is very essential to take the benefit of study of correlation. For this, we find there are various methods of measurement of correlation, which can be represented as given below



Among these methods we will discuss only Scatter Diagram, Karl Pearson's Coefficient of Correlation and Spearman's Rank Coefficient of Correlation.

3.1 Bivariate statistical distributions

We consider a population of N individuals measured simultaneously by the two characters X and Y which may be qualitative or quantitative, and which may not be of the same nature. The k modalities of X are denoted by $x_1, \dots, x_j, \dots, x_k$; the l modalities of Y are denoted by $y_1, \dots, y_j, \dots, y_l$.

3.1.1 Statistical table

The distribution of the N observations, or joint distribution, according to the modalities of X and Y is presented in the form of a double-entry table, called a **contingency** table or a double-entry table or a **cross table** or sometimes a correlation table (table of k rows and l columns).

$X \backslash Y$	y_1	y_2	$\dots$	y_j	$\dots$	y_l	TOTAL
x_1	n_{11}	n_{12}	$\dots$	n_{1j}	$\dots$	n_{1l}	$n_{1\cdot}$
x_2	n_{21}	n_{22}	$\dots$	n_{2j}	$\dots$	n_{2l}	$n_{2\cdot}$
$\cdot$	$\cdot$	$\cdot$		$\cdot$		$\cdot$	$\cdot$
$\cdot$	$\cdot$	$\cdot$	$\dots$	$\cdot$	$\dots$	$\cdot$	$\cdot$
$\cdot$	$\cdot$	$\cdot$		$\cdot$		$\cdot$	$\cdot$
x_i	n_{i1}	n_{i2}	$\dots$	n_{ij}	$\dots$	n_{il}	$n_{i\cdot}$
$\cdot$	$\cdot$	$\cdot$		$\cdot$		$\cdot$	$\cdot$
$\cdot$	$\cdot$	$\cdot$	$\dots$	$\cdot$	$\dots$	$\cdot$	$\cdot$
$\cdot$	$\cdot$	$\cdot$		$\cdot$		$\cdot$	$\cdot$
x_k	n_{k1}	n_{k2}	$\dots$	n_{kj}	$\dots$	n_{kl}	$n_{k\cdot}$
TOTAL	$n_{\cdot 1}$	$n_{\cdot 2}$	$\dots$	$n_{\cdot j}$	$\dots$	$n_{\cdot l}$	$n_{\cdot\cdot} = N$

- The number n_{ij} indicates the number of times the modality x_i of the variable X and the modality y_j of the variable Y were observed simultaneously.
- The number $n_{i\cdot}$, called the marginal number of X , represents the total number of observations of modality x_i of X , whatever the modality of Y .

$$n_{i\cdot} = \sum_{j=1}^l n_{ij}$$

- Similarly, the number $n_{\cdot j}$, called the marginal number of Y , is the total number of observations of the modality y_j of Y , whatever the modality of X .

$$n_{\cdot j} = \sum_{i=1}^k n_{ij}$$

Obviously, we have

$$\sum_{i=1}^k \sum_{j=1}^l n_{ij} = \sum_{j=1}^l n_{\cdot j} = \sum_{i=1}^k n_{i\cdot} = N.$$

The joint distribution can also be dened by the frequencies

$$f_{ij} = \frac{n_{ij}}{N}$$

Example. Consider the following two-dimensional statistical series of the pair (X, Y)

X/Y	-2	0	2	3	$n_{i.}$
2	3	4	0	6	13
3	4	3	3	2	12
4	2	3	3	2	10
$n_{.j}$	9	10	6	10	35

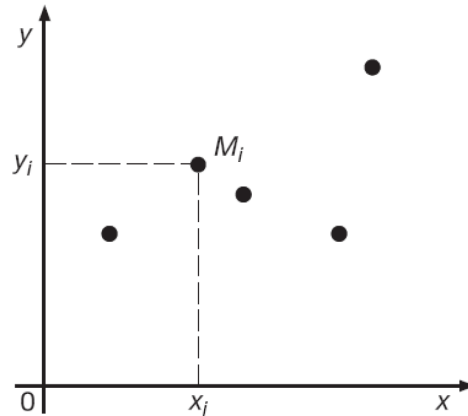
3.1.2 Graphical Representation

This is a very convenient graph for representing the simultaneous observations of two quantitative variables.

Scatter Diagram

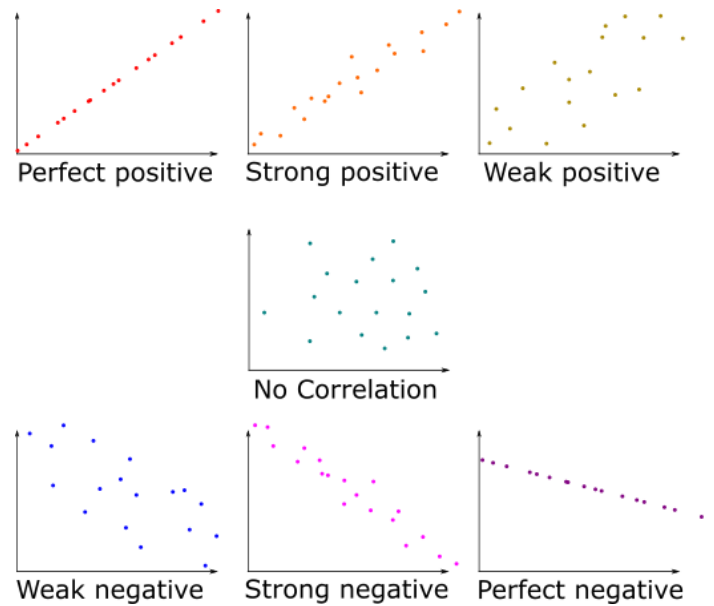
If the observations of two statistical variables X and Y are known individually, we start by visualizing them by representing them in the form of a cloud of points: in a Cartesian coordinate system, each observation $(x_i; y_i)$ is represented by the point M_i of coordinates $(x_i; y_i)$, and the shape of the cloud gives information on the type of a possible link.

This is graphic method of measurement of correlation. It is a diagrammatic representation of bivariate data to ascertain the relationship between two variables. Under this



method the given data are plotted on a graph paper in the form of dot. i.e. for each pair of X and Y values we put dots and thus obtain as many points as the number of observations. Usually an independent variable is shown on the X -axis whereas the dependent variable is shown on the Y -axis. Once the values are plotted on the graph it reveals the type of the correlation between variable X and Y . A scatter diagram reveals whether the movements in one series are associated with those in the other series.

- Perfect Positive Correlation: In this case, the points will form on a straight line falling from the lower left hand corner to the upper right hand corner.
- Perfect Negative Correlation: In this case, the points will form on a straight line rising from the upper left hand corner to the lower right hand corner.
- High Degree of Positive Correlation: In this case, the plotted points fall in a narrow band, wherein points show a rising tendency from the lower left hand corner to the upper right hand corner.
- High Degree of Negative Correlation: In this case, the plotted points fall in a narrow band, wherein points show a declining tendency from upper left hand



corner to the lower right hand corner.

- **Low Degree of Positive Correlation:** If the points are widely scattered over the diagrams, wherein points are rising from the left hand corner to the upper right hand corner.
- **Low Degree of Negative Correlation:** If the points are widely scattered over the diagrams, wherein points are declining from the upper left hand corner to the lower right hand corner.
- **Zero (No) Correlation:** When plotted points are scattered over the graph haphazardly, then it indicate that there is no correlation or zero correlation between two variables.

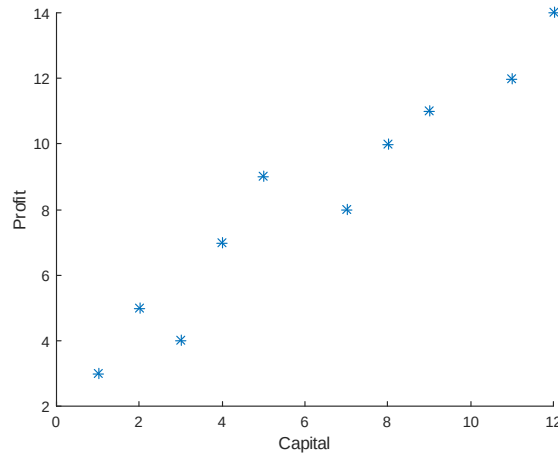
Example. Given the following pairs of values

Capital Employed (M)	1	2	3	4	5	7	8	9	11	12
Profit (M)	3	5	4	7	9	8	10	11	12	14

(a) Draw a scatter diagram

(b) Do you think that there is any correlation between profits and capital employed?

Is it positive or negative? Is it high or low?



From the observation of scatter diagram we can say that the variables are positively correlated. In the diagram the points trend toward upward rising from the lower left hand corner to the upper right hand corner, hence it is positive correlation. Plotted points are in narrow band which indicates that it is a case of high degree of positive correlation.

3.1.3 Marginal distributions

The marginal distribution is determined by isolating the first and last columns of the contingency table. The first column contains the modalities x_i and the last, the corresponding frequencies. That is to say on the margin of the contingency table, we can extract the data only with respect to X and only with respect to Y .

The k pairs $(x_i; n_i)$ form the marginal distribution of the variable X .

The l pairs $(y_j; n_j)$ form the marginal distribution of the variable Y .

Marginal distributions can also be given as frequencies

$$f_{i.} = \frac{n_{i.}}{N} \quad \text{and} \quad f_{.j} = \frac{n_{.j}}{N}.$$

Moreover, we have

$$\sum_{i=1}^k f_{i.} = \sum_{j=1}^l f_{.j} = 1.$$

These two distributions can be presented in the form of statistical tables

X	Marginal frequency	marginal relative frequency	Y	Marginal frequency	marginal relative frequency
x_1	$n_{1.}$	$f_{1.} = \frac{n_{1.}}{N}$	y_1	$n_{.1}$	$f_{.1} = \frac{n_{.1}}{N}$
x_2	$n_{2.}$	$f_{2.} = \frac{n_{2.}}{N}$	y_2	$n_{.2}$	$f_{.2} = \frac{n_{.2}}{N}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_k	$n_{k.}$	$f_{k.} = \frac{n_{k.}}{N}$	y_l	$n_{.l}$	$f_{.l} = \frac{n_{.l}}{N}$
Total	N	1	Total	N	1

3.1.4 Numerical description

Having a joint distribution, we can deduce the marginal distributions which allow us to study each variable separately by graphically representing its distribution and, if it is a quantitative variable, by calculating its central tendency and dispersion characteristics.

Characteristic of marginals

The marginal means of the variables X and Y are:

$$\bar{X} = \frac{1}{N} \sum_{i=1}^k n_{i.} x_i = \sum_{i=1}^k f_{i.} x_i \quad \text{and} \quad \bar{Y} = \frac{1}{N} \sum_{j=1}^l n_{.j} y_j = \sum_{j=1}^l f_{.j} y_j.$$

The marginal variances of the variables X and Y are given by :

$$Var(X) = \overline{X^2} - \overline{X}^2 = \sum_{i=1}^k f_{i.} x_i^2 - \left(\sum_{i=1}^k f_{i.} x_i \right)^2$$

and

$$Var(Y) = \overline{Y^2} - \overline{Y}^2 = \sum_{j=1}^l f_{.j} y_j^2 - \left(\sum_{j=1}^l f_{.j} y_j \right)^2$$

The marginal standard deviations of X and Y are given by the squared roots of their variances.

Example. Reconsider the following two-dimensional statistical series of the pair (X, Y)

X/Y	-2	0	2	3	$n_{i.}$
2	3	4	0	6	13
3	4	3	3	2	12
4	2	3	3	2	10
$n_{.j}$	9	10	6	10	35

$$\overline{X} = \frac{1}{N} \sum_{i=1}^k n_{i.} x_i = \frac{1}{35} (2 \times 13 + 3 \times 12 + 4 \times 10) = 2.9143.$$

and

$$\overline{Y} = \frac{1}{N} \sum_{j=1}^l n_{.j} y_j = \frac{1}{35} (-2 \times 9 + 0 \times 10 + 2 \times 6 + 3 \times 10) = 0.68571.$$

The marginal variances of the variables X and Y are

$$Var(X) = \overline{X^2} - \overline{X}^2 = \frac{1}{35} (2^2 \times 13 + 3^2 \times 12 + 4^2 \times 10) - (2.9143)^2 = 0.64971$$

and

$$Var(Y) = \overline{Y^2} - \overline{Y}^2 = \frac{1}{35} ((-2)^2 \times 9 + 0^2 \times 10 + 2^2 \times 6 + 3^2 \times 10) - (0.68571)^2 = 3.$$

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Conditional distributions

The distribution of the variable Y knowing the variable X being equal to x_i , is called the conditional distribution of Y for $X = x_i$:

$Y \mid X = x_i$	y_1	$\cdots$	y_j	$\cdots$	y_l	Total
Frequency	n_{i1}	$\cdots$	n_{ij}	$\cdots$	n_{il}	$n_{i.}$

This distribution of n_i observations, satisfying the condition $X = x_i$, is presented in the form of conditional relative frequencies:

$$f_{j/i} = \frac{n_{ij}}{n_{i.}} \quad \text{and} \quad \sum_{j=1}^l f_{j/i} = 1.$$

There are k conditional distributions of Y for $i = 1, \dots, k$.

When the variable Y is quantitative, we can calculate for each value x_i its conditional mean $\bar{Y}_i$ and its conditional variance

$$\bar{Y}_i = \sum_{j=1}^l f_{j/i} y_j \quad \text{and} \quad \text{Var}(Y \mid X = x_i) = \sum_{j=1}^l f_{j/i} (y_j - \bar{Y}_i)^2$$

The k modalities of X inducing a partition of the observations into k subgroups, the mean can be expressed as a weighted sum of the k means $\bar{Y}_i$

$$\bar{Y} = \sum_{i=1}^k f_{i.} \bar{Y}_i$$

Symmetrically, we have l conditional distributions of X and we define the conditional relative frequencies $f_{i/j}$

$$f_{i/j} = \frac{n_{ij}}{n_{.j}} \quad \text{and} \quad \sum_{i=1}^k f_{i/j} = 1.$$

$X \mid Y = y_j$	x_1	$\cdots$	x_i	$\cdots$	x_k	Total
Frequency	$f_{1/j}$	$\cdots$	$f_{i/j}$	$\cdots$	$f_{k/j}$	1

When the variable X is quantitative, we can calculate for each value y_j its conditional mean $\bar{X}_j$ and its conditional variance

$$\bar{X}_j = \sum_{i=1}^k f_{i/j} x_i \quad \text{and} \quad \text{Var}(X \mid Y = y_j) = \sum_{i=1}^k f_{i/j} (x_i - \bar{X}_j)^2$$

We have the following relationship between the mean $\bar{X}$ and the l conditional means $\bar{X}_j$

$$\bar{X} = \sum_{j=1}^l f_{.j} \bar{X}_j.$$

Example. Retake the previous example. thus, to determine the conditional mean of X when $Y = 2$, it suffices to observe the behavior of X relative to the column $Y = 2$.

X	n_{i2}
2	0
3	3
4	3
$n_{.2}$	6

$$\bar{X}_2 = \frac{1}{6} (0 \times 2 + 3 \times 3 + 4 \times 3) = 3.5$$

3.1.5 Covariance between two statistical variables

The covariance is equal to the average of the deviations of the pairs $(x_i; y_i)$ of X and Y with respect to the point $(\bar{X}; \bar{Y})$

$$\text{Cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{X}) (y_i - \bar{Y}) = \frac{1}{N} \sum_{i=1}^N x_i y_i - \bar{X} \bar{Y}$$

In the case of grouped data in a contingency table (weighted covariance) is given by

$$Cov(X, Y) = \frac{1}{N} \sum_{i=1}^k \sum_{j=1}^l (x_i - \bar{X}) (y_j - \bar{Y}) = \frac{1}{N} \sum_{i=1}^k \sum_{j=1}^l n_{ij} x_i y_j - \bar{X} \bar{Y}$$

The covariance indicates the direction of the relationship between the variables X and Y .

Thus, the following cases can be distinguished:

- If $Cov(X; Y) > 0$, then we can say that the relationship between the two variables is positive.

In this case, these two variables vary in the same direction.

- If $Cov(X; Y) < 0$; then we can say that the relationship between the two variables is negative.

In this case, these two variables vary in opposite directions.

- If $Cov(X; Y) = 0$, then we can say that there is no relationship between the two variables.

In this case, the variations of one do not lead to the variation of the other.

Covariance properties

1. $Cov(X; Y) = Cov(Y; X)$.
2. $Cov(X; X) = Var(X)$.
3. $Var(X + Y) = var(X) + var(Y) + 2cov(X; Y)$.
4. $\forall a; b; x_0; y_0 \in \mathbb{R} : Cov(aX + x_0; bY + y_0) = abCov(X; Y) \implies Var(aX + bY + c) = a^2Var(X) + b^2Var(Y) + 2abCov(X; Y)$.
5. $|Cov(X; Y)| \leq \sqrt{Var(X)Var(Y)}$.

The magnitude of the covariance is not very informative since it is affected by the magnitude of both X and Y . However, the sign of the covariance tells us something useful about the relationship between X and Y .

Consider the following conditions:

- $x_i > \bar{X}$ and $y_j > \bar{Y}$ then $(x_i - \bar{X})(y_j - \bar{Y})$ will be positive.
- $x_i < \bar{X}$ and $y_j < \bar{Y}$ then $(x_i - \bar{X})(y_j - \bar{Y})$ will be positive.
- $x_i > \bar{X}$ and $y_j < \bar{Y}$ then $(x_i - \bar{X})(y_j - \bar{Y})$ will be negative.
- $x_i < \bar{X}$ and $y_j > \bar{Y}$ then $(x_i - \bar{X})(y_j - \bar{Y})$ will be negative.

Since $Cov(X, Y)$ depends on the magnitude of X and Y we would prefer to have a measure of association that is not affected by changes in the scales of the variables.

The most common measure of linear association is correlation where the magnitude of the correlation measures the strength of the linear association and the sign determines if it is a positive or negative relationship.

3.1.6 Karl Pearson's Coefficient of linear Correlation

Karl Pearson's method of calculating coefficient of correlation is based on the covariance of the two variables in a series. This method is widely used in practice and the coefficient of correlation is denoted by the symbol " ρ ". If the two variables under study are X and Y , the following formula suggested by Karl Pearson can be used for measuring the degree of relationship of correlation.

$$\rho(X, Y) = \frac{Cov(X, Y)}{\sigma_X \cdot \sigma_Y}$$

$$-1 \leq \rho(X, Y) \leq 1$$

Remarque 3.1.1 *The link between two numerical variables can be studied by the correlation coefficient. Nevertheless, it should be kept in mind that the Pearson correlation coefficient only measures linear relationships, and its value is in no way a reflection of the existence of a causal link between the two variables. Given random variables X and Y*

$$X \text{ and } Y \text{ are independent} \implies \text{Cov}(X, Y) = \rho(X, Y) = 0$$

$$\text{Cov}(X, Y) = \rho(X, Y) = 0 \not\Rightarrow X \text{ and } Y \text{ are independent}$$

Properties of the linear correlation coefficient:

1. The correlation coefficient is always between -1 and +1.
2. If $\rho = +1$ then the points are all on the same increasing line, the perfect positive linear correlation.
3. If $\rho = -1$ then the points are all on the same decreasing line, the perfect negative linear correlation.
4. If $\rho = 0$ then there is no linear relationship between the variables X and Y .
5. We have for all $a, b, x_0, y_0 \in \mathbb{R}$:

$$\begin{aligned} \rho(aX + x_0, bY + y_0) &= \frac{\text{Cov}(aX + x_0, bY + y_0)}{\sigma_{aX+x_0} \cdot \sigma_{bY+y_0}} = \frac{ab\text{Cov}(X, Y)}{|ab| \sigma_X \cdot \sigma_Y} \\ &= \begin{cases} \rho(X, Y) & \text{if } ab > 0 \\ -\rho(X, Y) & \text{if } ab < 0 \end{cases} \end{aligned}$$

Example. From following information find the correlation coefficient between advertisement expenses and sales volume using Karl Pearson's coefficient of correlation method.

Firm	1	2	3	4	5	6	7	8	9	10
Advertisement Exp.	11	13	14	16	16	15	15	14	13	13
Sales Volume	50	50	55	60	65	65	65	60	60	50

Let us assume that advertisement expenses are variable X and sales volume are variable Y .

Firm	x_i	y_i	$(x_i - \bar{X})^2$	$(y_i - \bar{Y})^2$	$(x_i - \bar{X})(y_i - \bar{Y})$
1	11	50	9	64	24
2	13	50	1	64	8
3	14	55	0	9	0
4	16	60	4	4	4
5	16	65	4	49	14
6	15	65	1	49	7
7	15	65	1	49	7
8	14	60	0	4	0
9	13	60	1	4	-2
10	13	50	1	64	8
Σ	140	580	22	360	70

$$\bar{X} = \frac{140}{10} = 10 \quad \text{and} \quad \bar{Y} = \frac{580}{10} = 58$$

$$\rho(X, Y) = \frac{\sum (x_i - \bar{X})(y_i - \bar{Y})}{\sqrt{\sum (x_i - \bar{X})^2} \times \sqrt{\sum (y_i - \bar{Y})^2}} = \frac{\mathbf{70}}{\sqrt{22} \times \sqrt{360}} = 0.78657.$$

Interpretation: From the above calculation it is very clear that there is high degree of positive correlation i.e. $\rho = 0.7866$, between the two variables. i.e. Increase in advertisement expenses leads to increased sales volume.

Example. Find the correlation coefficient between age and playing habits of the following students using Karl Pearson's coefficient of correlation method.

Age	15	16	17	18	19	20
Number of students	250	200	150	120	100	80
Regular Players	200	150	90	48	30	12

To find the correlation between age and playing habits of the students, we need to compute the percentages of students who are having the playing habit.

$$\text{Percentage of playing habits} = (\text{No. of Regular Players} / \text{Total No. of Students}) \times 100$$

Now, let us assume that ages of the students are variable X and percentages of playing habits are variable Y .

Age (X)	% of playing habits (Y)	$(x_i - \bar{X})^2$	$(y_i - \bar{Y})^2$	$(x_i - \bar{X})(y_i - \bar{Y})$
15	$200/250 \times 100 = 80$	6.25	900	-75
16	75	2.25	625	-37.5
17	60	0.25	100	-5
18	40	0.25	100	-5
19	30	2.25	400	-30
20	15	6.25	1225	-87.5
Σ	105	300	3350	-240

$$\bar{X} = \frac{105}{6} = 17.5 \quad \text{and} \quad \bar{Y} = \frac{300}{6} = 50$$

$$\rho(X, Y) = \frac{\sum (x_i - \bar{X})(y_i - \bar{Y})}{\sqrt{\sum (x_i - \bar{X})^2} \times \sqrt{\sum (y_i - \bar{Y})^2}} = \frac{-240}{\sqrt{17.5 \times 3350}} = -0.99122$$

Interpretation: From the above calculation it is very clear that there is high degree of negative correlation i.e. $\rho(X, Y) = -0.9912$, between the two variables of age and playing habits. i.e. Playing habits among students decreases when their age increases.

3.1.7 Spearman's Rank Coefficient of Correlation:

When quantification of variables becomes difficult such leadership ability, knowledge of person etc, then this method of rank correlation is useful which was developed by British psychologist Charles Edward Spearman in 1904. In this method ranks are allotted to each element either in ascending or descending order.

The correlation coefficient between these allotted two series of ranks is popularly called as "Spearman's Rank Correlation" and denoted by " r_s ". It is defined as the Pearson correlation coefficient between the rank variables. For a sample of size N , the N raw scores x_i, y_i are converted to ranks $R(x_i), R(y_i)$ and r_s is computed as

$$r_s = \rho(R(X), R(Y)) = \frac{Cov(R(X), R(Y))}{\sigma_{R(X)}\sigma_{R(Y)}}$$

where ρ denotes the usual Pearson correlation coefficient, but applied to the rank variables, $Cov(R(X), R(Y))$ is the covariance of the rank variables, $\sigma_{R(X)}$ and $\sigma_{R(Y)}$ are the standard deviations of the rank variables. Only if all N ranks are distinct

integers, it can be computed using the popular formula

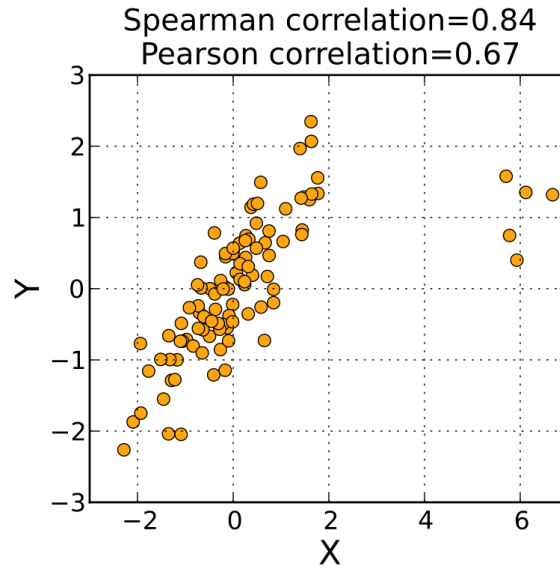
$$r_s = 1 - \frac{6 \sum d_i^2}{N(N^2 - 1)}$$

where, $d_i = R(x_i) - R(y_i)$ is the difference between the two ranks of each observation, N = Number of pairs of ranks.

Important Inference to keep in mind: The Spearman correlation can evaluate a monotonic relationship between two variables — Continuous or Ordinal and it is based on the ranked values for each variable rather than the raw data.

In case of tie in ranks or equal ranks

In some cases it may be possible that it becomes necessary to assign same rank to two or more elements or individual or entries. In such situation, it is customary to give each individual or entry an average rank. For example, if two individuals are ranked equal to 5th place, then both of them are allotted with common rank $(5 + 6)/2 = 5.5$ and if three are ranked in 5th place, then they are given the rank of $(5 + 6 + 7)/3 = 6$. It means where two or more individuals are to be ranked equal, the rank assigned for the purpose of calculating coefficient of correlation is the average of the ranks with these individual or items or entries would have got had they differed slightly with each other. Where equal ranks are assigned to some entries, an adjustment factor is to be added to the value of $6 \sum d_i^2$ in the above formula for calculating the rank coefficient correlation. This adjustment factor is to be added for every repetition of rank. Adjustment factor $= \frac{m \times (m^2 - 1)}{12}$ where, m = number of items whose rank are common. For example, if a particular rank repeated two times then $m=2$ and if it repeats three times then $m = 3$ and so on.



Hence the above formula can be re-written as follows:

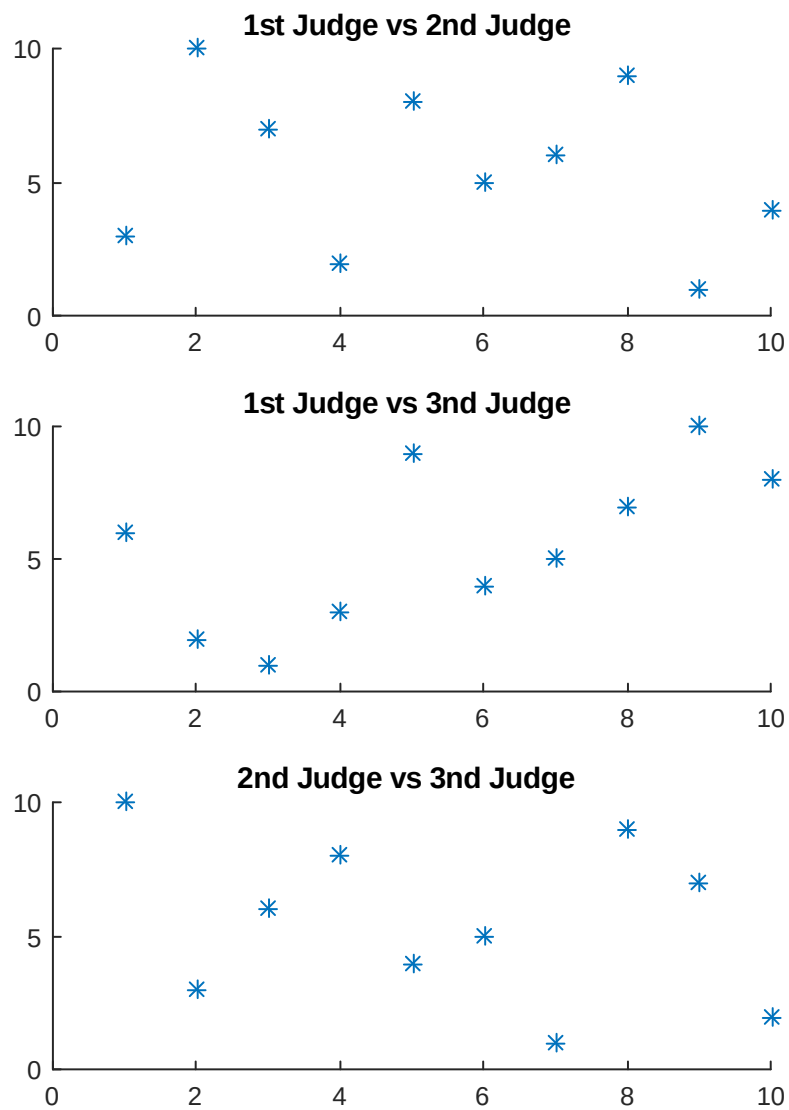
$$r_s = 1 - \frac{6 \left[\sum d_i^2 + \frac{m_1 \times (m_1^2 - 1)}{12} + \frac{m_2 \times (m_2^2 - 1)}{12} + \dots \right]}{N(N^2 - 1)}$$

The Spearman correlation is less sensitive than the Pearson correlation to strong outliers that are in the tails of both samples. That is because Spearman's r_s limits the outlier to the value of its rank.

Example. Ten competitors in a skill contest are ranked by three judges in the following order:

1st Judge	1	6	5	10	3	2	4	9	7	8
2nd Judge	3	5	8	4	7	10	2	1	6	9
3rd Judge	6	4	9	8	1	2	3	10	5	7

Use the rank correlation coefficient to determine which pairs of judges has the nearest approach to common tastes.



In order to find out which pair of judges has the nearest approach to common tastes, we compare rank correlation between the judgements of

1. 1st Judge and 2nd Judge
2. 2nd Judge and 3rd Judge
3. 1st Judge and 3rd Judge

Note $R1$ =Rank by 1st Judge, $R2$ =Rank by 2nd Judge, $R3$ =Rank by 3rd Judge.

$R1$	$R2$	$R3$	$d^2 = (R1 - R2)^2$	$d^2 = (R2 - R3)^2$	$d^2 = (R1 - R3)^2$
1	3	6	4	9	25
6	5	4	1	1	4
5	8	9	9	1	16
10	4	8	36	16	4
3	7	1	16	36	4
2	10	2	64	64	0
4	2	3	4	1	1
9	1	10	64	81	1
7	6	5	1	1	4
8	9	7	1	4	1
			$\sum d^2 = 200$	$\sum d^2 = 214$	$\sum d^2 = 60$

1. 1st Judge and 2nd Judge: $r_s = 1 - \frac{6 \sum d_i^2}{N(N^2-1)} = 1 - \frac{6 \times 200}{10 \times (10^2-1)} = -0.212\ 12$
2. 2nd Judge and 3rd Judge: $r_s = 1 - \frac{6 \sum d_i^2}{N(N^2-1)} = 1 - \frac{6 \times 214}{10 \times (10^2-1)} = -0.296\ 97$
3. 1st Judge and 3rd Judge: $r_s = 1 - \frac{6 \sum d_i^2}{N(N^2-1)} = 1 - \frac{6 \times 60}{10 \times (10^2-1)} = 0.636\ 36$.

Interpretation: From the above calculation it can be observed that coefficient of correlation is only positive in the judgement of the first and third judges. Therefore, it can be concluded that first and third judges have the nearest approach to common

tastes.

Example. From the following data, compute the rank correlation.

X	82	68	75	61	68	73	85	68
Y	81	71	71	68	62	69	80	70

Calculation of Spearman's Rank Coefficient of Correlation

X	Y	R_1	R_2	$d^2 = (R_1 - R_2)^2$
82	81	2	1	1
68	71	6	3.5	6.25
75	71	3	3.5	0.25
61	68	8	7	1
68	62	6	8	4
73	69	4	6	4
85	80	1	2	1
68	70	6	5	1
				$\sum d^2 = 18.5$

In the problem we find there are repetitions of ranks. Value of $X = 68$ repeated 3 times and value of $Y = 71$ repeated 2 times. Therefore we need to compute adjustment factor to be added to the value of $\sum d^2$. We have

$$r_s = 1 - \frac{6 \left[\sum d_i^2 + \frac{m_1 \times (m_1^2 - 1)}{12} + \frac{m_2 \times (m_2^2 - 1)}{12} \right]}{N(N^2 - 1)}$$

where m_1 is for the value X repeated three times, $m_1 = 3$ and for value Y repeated two times, $m_2 = 2$. Thus

$$r_s = 1 - \frac{6 \times \left[18.5 + \frac{3 \times (3^2 - 1)}{12} + \frac{2 \times (2^2 - 1)}{12} \right]}{8 \times (8^2 - 1)} = 0.75.$$

Spearman's Rank Coefficient of Correlation = 0.75, which indicates there is high degree of positive correlation.

3.2 Regression Analysis

Regression analysis is one of the most commonly used statistical techniques in social and behavioral sciences as well as in physical sciences which involves identifying and evaluating the relationship between a dependent variable and one or more independent variables, which are also called predictor or explanatory variables. It is particularly useful for assess and adjusting for confounding. Model of the relationship is hypothesized and estimates of the parameter values are used to develop an estimated regression equation. Various tests are then employed to determine if the model is satisfactory. If the model is deemed satisfactory, the estimated regression equation can be used to predict the value of the dependent variable given values for the independent variables. Linear regression explores relationships that can be readily described by straight lines or their generalization to many dimensions. A surprisingly large number of problems can be solved by linear regression, and even more by means of transformation of the original variables that result in linear relationships among the transformed variables. When there is a single continuous dependent variable and a single independent variable, the analysis is called a **simple linear regression analysis**. This analysis assumes that there is a linear association between the two variables. **Multiple regression** is to learn more about the relationship between several independent or predictor variables and a dependent or criterion variable.

Independent variables are characteristics that can be measured directly; these variables are also called predictor or explanatory variables used to predict or to explain the

behavior of the dependent variable.

Dependent variable is a characteristic whose value depends on the values of independent variables.

Objectives of Regression Analysis

Regression analysis used to explain variability in dependent variable by means of one or more of independent or control variables and to analyze relationships among variables to answer; the question of how much dependent variable changes with changes in each of the independent's variables, and to forecast or predict the value of dependent variable based on the values of the independent's variables.

The primary objective of regression is to develop a linear relationship between a response variable and explanatory variables for the purposes of prediction, assumes that a functional linear relationship exists, and alternative approaches (functional regression) are superior.

3.2.1 Assumption of Regression Analysis

The regression model is based on the following assumptions.

- The relationship between independent variable and dependent is linear.
- The expected value of the error term is zero
- The variance of the error term is constant for all the values of the independent variable, the assumption of homoscedasticity.
- There is no autocorrelation.
- The independent variable is uncorrelated with the error term.

- The error term is normally distributed.
- On an average difference between the observed value and the predicted value is zero.
- On an average the estimated values of errors and values of independent variables are not related to each other.
- The squared differences between the observed value and the predicted value are similar.
- There is some variation in independent variable. If there are more than one variable in the equation, then two variables should not be perfectly correlated.

3.2.2 Simple Regression Model

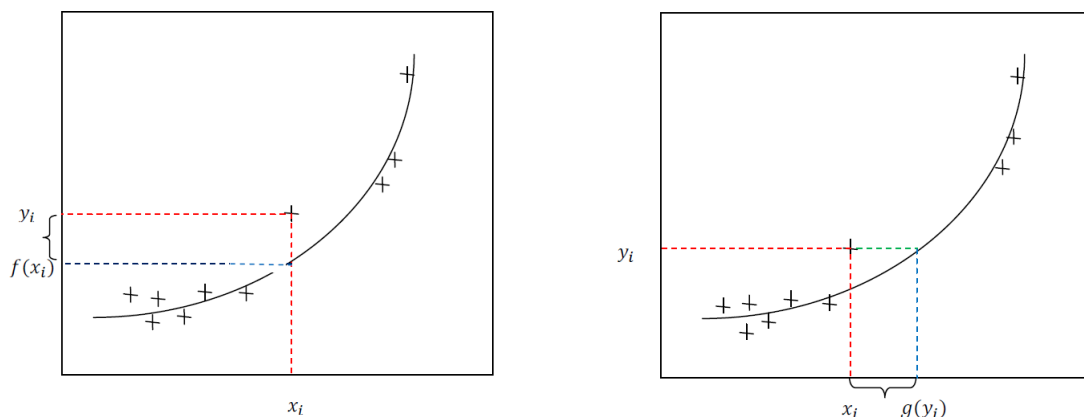
Simple linear regression is a statistical method that allows us to summarize and study relationships between two continuous (quantitative) variables. In a cause and effect relationship, the independent variable is the cause, and the dependent variable is the effect. Least squares linear regression is a method for predicting the value of a dependent variable y , based on the value of an independent variable x .

- One variable, denoted (x) , is regarded as the predictor, explanatory, or independent variable.
- The other variable, denoted (y) , is regarded as the response, outcome, or dependent variable.

We choose a mathematical function whose graphical representation approaches the shape of the point cloud as closely as possible. The problem is to determine the coefficients of the function which minimize the sum of the squares of the deviations between

the points representative of the observations and the curve representative of the function.

In the least squares method, the deviations are measured parallel to the axes, that is to say vertically or horizontally in an orthogonal frame.



Differences measured vertically (Y in X) Differences measured horizontally (X in Y)

The fitted curve of Y on X

Let $(x_i, y_i), i = 1, \dots, n$, n observations of the pair (X, Y) . We are looking for a function $f : X \longrightarrow Y = f(X)$. Let $f(x_i)$ be the value taken by the function f when $X = x_i$. The vertical difference between the point (x_i, y_i) of the series and the curve is noted $\varepsilon_i = y_i - f(x_i)$. The gap ε_i is called residual gap or residual. The sum of the squared deviations calculated for each point of the cloud is equal to $\sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (y_i - f(x_i))^2$.

We call The fitted curve of Y on X and we note $C_{Y/X}$, the representative curve of the function such that the sum $\sum_{i=1}^n (y_i - f(x_i))^2$ be minimal.

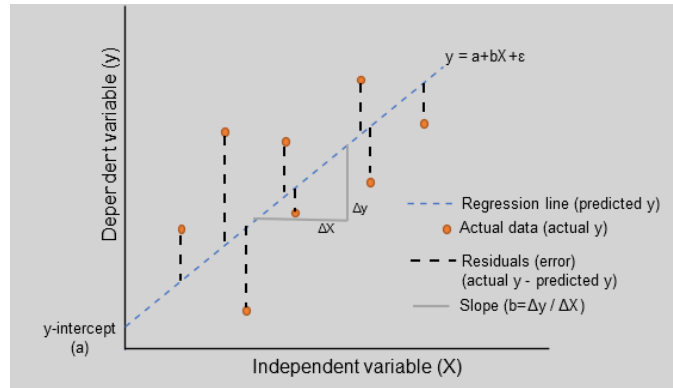
The fitted curve of X on Y

We are looking for a function $g : Y \longrightarrow X = g(Y)$. Let $g(y_i)$ the value taken by the function g when $Y = y_i$. The horizontal difference between the point (x_i, y_i) of the series and the curve is written $\varepsilon'_i = x_i - g(y_i)$. The sum that must be made minimal is in this case is $\sum_{i=1}^n \varepsilon_i'^2 = \sum_{i=1}^n (x_i - g(y_i))^2$.

Remarque 3.2.1 *The least squares method provides two curves of fit for a given statistical series. The two curves are all the closer to each other as the dispersion of the points is low.*

Linear fitting

When the shape of the point cloud leads to retaining the adjustment of a straight line, we say that we are making a linear fitting.



The fitted line of Y on X

The fitted line has an equation of the form $y = ax + b$. We put $\hat{y}_i = ax_i + b$ the estimated value of y_i by the linear model. The difference between the point (x_i, y_i) of the cloud and the line is then $\varepsilon_i = y_i - ax_i - b$.

We call the fitted line of Y on X or the regression line of Y on X , and we note it $D_{Y/X}$, the line of equation $y = ax + b$ such that the sum $\sum_{i=1}^n (y_i - \hat{y}_i)^2$ is minimal.

To find the optimal line in the sense of least squares, it is necessary to find the coefficients a (slope) and b (intercept).

Note

$$\phi(a, b) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - ax_i - b)^2$$

To find the optimal line, we must solve the system of partial derivatives of the first order

$$\begin{cases} \frac{\partial \phi(a, b)}{\partial a} = 0 \\ \frac{\partial \phi(a, b)}{\partial b} = 0 \end{cases}$$

which gives

$$\begin{cases} \sum_{i=1}^n -2x_i (y_i - ax_i - b) = 0 \\ \sum_{i=1}^n -2 (y_i - ax_i - b) = 0 \end{cases} \iff \begin{cases} \sum_{i=1}^n x_i y_i - a \sum_{i=1}^n x_i^2 - b \sum_{i=1}^n x_i = 0 \\ \sum_{i=1}^n y_i - a \sum_{i=1}^n x_i - nb = 0 \end{cases}$$

The solution to this system of equations is given by

$$a = \frac{Cov(X, Y)}{Var(X)}$$

$$b = \bar{Y} - a\bar{X}$$

The fitted line of X on Y

The fitted line has an equation of the form $x = a'y + b'$. We put $\hat{x}_i = a'y_i + b'$ the estimated value of x_i by the linear model. The difference between the point (x_i, y_i) of the cloud and the line is then $\varepsilon'_i = x_i - a'y_i - b'$.

We call the fitted line of X on Y or the regression line of X on Y , and we note it

$D_{X/Y}$, the line of equation $x = a'y + b'$ such that the sum $\sum_{i=1}^n (x_i - \hat{x}_i)^2$ is minimal.

To find the optimal line in the sense of least squares, it is necessary to find the coefficients a (slope) and b (intercept).

Note

$$\psi(a', b') = \sum_{i=1}^n (x_i - \hat{x}_i)^2 = \sum_{i=1}^n (x_i - a'y_i - b')^2$$

The values of a' and b' are given by

$$a' = \frac{\text{Cov}(X, Y)}{\text{Var}(Y)}$$

$$b = \bar{X} - a'\bar{Y}$$

Remarque 3.2.2 • *The two regression lines $D_{Y/X}$ and $D_{X/Y}$ are generally distinct. They intersect at the center of gravity $G(\bar{X}, \bar{Y})$ of the cloud. The two slopes a and a' always have the same sign, that of the covariance.*

- *To draw the two regression lines on the same graph, we write $D_{X/Y}$ as follows :*

$$D_{X/Y} : y = \frac{x}{a'} - \frac{b'}{a'}$$

- *When the two variables are independent, the two regression lines $D_{Y/X}$ and $D_{X/Y}$ are perpendicular and therefore $aa' = 0$.*
- *When the variables X and Y are functionally related, the two regression lines coincide and therefore $aa' = a \frac{1}{a} = 1$*
- *In general , $0 \leq aa' \leq 1$.*

Example. (Ungrouped Data) Research is carried out on the speed of propagation of

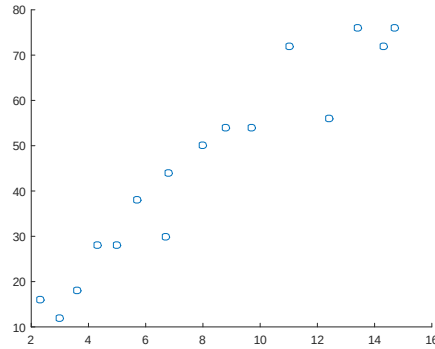
nerve impulses in a nerve fiber. We denote by X the diameter in microns of nerve fibers and Y the speed in meters per second of the nerve impulse in the fiber of diameter X .

The results are given in the table below

X	2.3	3	3.6	4.3	5	5.7	6.7	6.8	8	8.8	9.7	11	12.4	13.4	14.3	14.7
Y	16	12	18	28	28	38	30	44	50	54	54	72	56	76	72	76

We give $\sum_{i=1}^n x_i = 129.7$, $\sum_{i=1}^n y_i = 724$, $\sum_{i=1}^n x_i^2 = 1304.79$, $\sum_{i=1}^n y_i^2 = 39960$, $\sum_{i=1}^n x_i y_i = 7165.4$.

1. Represent the scatterplot associated with this double statistical series.



2. Calculate the means of each of the variables X and Y .

$$\bar{X} = \frac{1}{16} \sum_{i=1}^n x_i = \frac{129.7}{16} = 8.1063$$

$$\bar{Y} = \frac{1}{16} \sum_{i=1}^n y_i = \frac{724}{16} = 45.25$$

3. Calculate the variances of each of the variables.

$$\sigma_X^2 = \frac{1}{16} \sum_{i=1}^n x_i^2 - \bar{X}^2 = \frac{1304.79}{16} - (8.1063)^2 = 15.837$$

$$\sigma_Y^2 = \frac{1}{16} \sum_{i=1}^n y_i^2 - \bar{Y}^2 = \frac{39960}{16} - (45.25)^2 = 449.94$$

4. Calculate the covariance between X and Y

$$Cov(X, Y) = \frac{1}{16} \sum_{i=1}^n x_i y_i - \bar{X} \bar{Y} = \frac{7165.4}{16} - 8.1063 \times 45.25 = 81.027$$

5. Find the equations of the regression lines of Y on X and of X on Y .

$$a = \frac{Cov(X, Y)}{Var(X)} = \frac{81.027}{15.837} = 5.1163$$

$$b = \bar{Y} - a\bar{X} = 45.25 - 5.1163 \times 8.1063 = 3.7757$$

which gives

$$D_{Y/X} : y = 5.1163x + 3.7757.$$

Also, for $D_{X/Y}$ we have

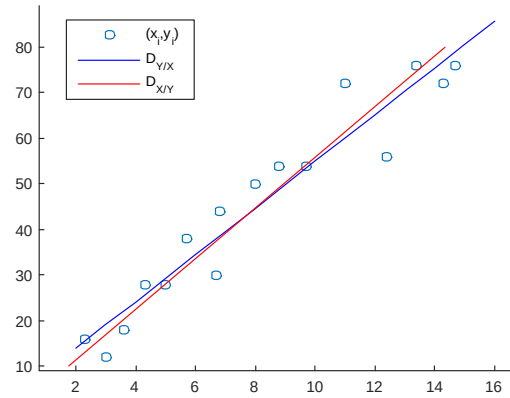
$$a' = \frac{Cov(X, Y)}{Var(Y)} = \frac{81.027}{449.94} = 0.18008$$

$$b = \bar{X} - a'\bar{Y} = 8.1063 - 0.18008 \times 45.25 = -0.04232$$

and

$$D_{X/Y} : x = 0.18008y - 0.04232.$$

6. Draw the two regression lines



If we look for the diameter of a nerve fiber through which the nerve impulse would propagate at the speed of $100m/s$ then we use the regression lines $D_{X/Y}$ to predict its value

$$x = 0.180\,08 \times 100 - 0.042\,32 = 17.966 \text{ microns}$$

and to determine the speed of the nerve impulse through a nerve fiber with a diameter of 18 microns, we use the regression lines $D_{Y/X}$

$$y = 5.116\,3 \times 18 + 3.775\,7 = 95.869m/s$$

Example. (Grouped Data) The table below represents the distribution of 28 students according to the annual number of absences (X) and the final grade (Y).

$X \backslash Y$	$[0,5[$	$[5,10[$	$[10,15[$	$[15,20[$
0	2	3	3	0
1	0	1	2	3
2	0	0	1	1
3	4	3	0	0
4	1	0	4	0

Give the equation of the regression lines of Y on X and that of X on Y .

$x_i \backslash y_i$	2.5	7.5	12.5	17.5	$n_{i.}$	$n_{i.}x_i$	$n_{i.}x_i^2$
0	2 $n_{11}x_1y_1 = 0$	3 0	3 0	0 0	8	0	0
1	0 0	1 7.5	2 25	3 52.5	6	6	6
2	0 0	0 0	1 25	1 35	2	4	8
3	4 30	3 67.5	0 0	0 0	7	21	63
4	1 10	0 0	4 200	0 0	5	20	80
$n_{.j}$	7	7	10	4	28	$\sum = 51$	$\sum = 157$
$n_{.j}y_j$	17.5	52.5	125	70	$\sum = 265$		
$n_{.j}y_j^2$	43.75	393.75	1562.5	1225	$\sum = 3225$		

$$\bar{X} = \frac{1}{28} \sum n_{i.}x_i = 1.8214 \quad \bar{Y} = \frac{1}{28} \sum n_{.j}y_j = 9.4642$$

$$\sigma_X^2 = \frac{1}{28} \sum n_{i.}x_i^2 - \bar{X}^2 = 2.289541 \quad \sigma_Y^2 = \frac{1}{28} \sum n_{.j}y_j^2 - \bar{Y}^2 = 25.60587$$

$$Cov(X, Y) = \frac{1}{28} \sum \sum n_{ij}x_iy_j - \bar{X}.\bar{Y} = -1.077806.$$

We obtain $a = \frac{Cov(X, Y)}{\sigma_X^2} = -0.470752$ and $b = \bar{Y} - a\bar{X} = 10.321731$, thus

$$D_{Y/X} : y = -0.470752x + 10.321731.$$

Also $a' = \frac{Cov(X, Y)}{\sigma_Y^2} = -0.04209215$ and $b' = \bar{X} - a'\bar{Y} = 2.219801$,

$$D_{X/Y} : x = -0.04209215y + 2.219801.$$

Residual variance and explained variance by a regression line

Fitted line of Y on X .

The quality of the adjustment is even better than the sum of the squares of the residuals (or deviations) between the line and the points of the series is small.

- We call residual variance of Y or residual variance of $D_{Y/X}$, and we note $V_R(Y)$, the following expression

$$V_R(Y) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- We call variance explained by the regression line $D_{Y/X}$, and we note $V_E(Y)$, the expression defined by

$$V_E(Y) = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - \bar{Y})^2.$$

Properties.

- $Var(Y) = V_R(Y) + V_E(Y)$.
- $V_E(Y) = \frac{Cov(X,Y)^2}{Var(X)} = Var(Y)\rho(X, Y)^2$.
- $V_R(Y) = Var(Y)(1 - \rho(X, Y)^2)$.

Coefficient of determination

We call the coefficient of determination the quantity R^2 defined by

$$R^2 = \frac{V_E(Y)}{Var(Y)} = \frac{V_E(X)}{Var(X)}$$

The coefficient of determination R^2 summarizes the part of the variance explained by the regression lines. It is a unitless number, between 0 and 1. It can be expressed as a percentage (percentage that the explained variance represents in relation to the total variance).

From $Var(Y) = V_R(Y) + V_E(Y)$, we have

$$R^2 = 1 - \frac{V_R(Y)}{Var(Y)} = 1 - \frac{V_R(X)}{Var(X)}.$$

R^2 is close to 1 if the points of the cloud are little dispersed around the regression line, it is close to 0 otherwise.

Example. from the previous example of nerve impulse propagation speed in a nerve fiber and diameter of a nerve fiber, we compute the coefficient of correlation

$$\rho(X, Y) = \frac{Cov(X, Y)}{\sigma_X \sigma_Y} = \frac{81.02969}{\sqrt{15.83809} \sqrt{449.9375}} = 0.9598794.$$

Calculate the residual variance of Y and the variance explained by the line $D_{Y/X}$

$$V_R(Y) = Var(Y)(1 - \rho(X, Y)^2) = 35.37928,$$

$$V_E(Y) = Var(Y)\rho(X, Y)^2 = 414.5582.$$

then we calculate the coefficient of determination

$$R^2 = \frac{V_E(Y)}{Var(Y)} = \frac{414.5582}{449.9375} = 0.92137$$

The regression lines explain 92.13% of the variance (of X and Y) which gives that the goodness of fit is very good.